

Abstract

This paper considers the relationship between market composition and temporal dynamics of ethno-racial stratification in mass-participatory asset bubbles. Beginning from the insight that bubbles possess the social structure of a Ponzi scheme, we first ask to what extent timing of entry is structured by racial status and associated resource disparities among participants. We then consider to what extent homophilous social cues and segregation amplify disparate rates of late-stage entry. We focus on the 2000-2008 U.S. residential housing market, using neighborhood-level data from the Home Mortgage Disclosure Act and the Federal Housing Finance Authority. Conditional on mortgage lending terms, credit availability, and other tract-level characteristics, trends in housing purchase rates for Blacks and Hispanics diverged from those of whites as prices neared their peaks. Residential segregation and processes of homophilous diffusion amplify price-bidding and exacerbate racial disparities in rates of late-stage entry by rendering actors less attuned to the emergent risks of buying at elevated prices, and more vulnerable to the overtures of predatory agents. The analysis links the sociology of financial markets to studies of racialized predatory inclusion by considering heterogenous market timing, a mechanism which has become increasingly salient as financialization draws more diverse actors into volatile asset markets.

Introduction

An oft-repeated legend in American financial history recounts that in 1929, just before the great stock market crash, billionaire John D. Rockefeller received a friendly stock tip from the man who was shining his shoes. Convinced it was a sign that the euphoria in the stock market had become unsustainable, he pulled the bulk of his money out of the market just before prices plummeted. The class-encoded moral of the story, as Davis (2009:214) puts it, is that “when one’s shoe-shin[er] is handing out stock tips, the smart money knows it is time to get out.”¹ The further sociological implication is that during periods of mass-participatory market exuberance, lower-status actors are disproportionately enticed into the market at late stages, when prices have nearly peaked and are primed to decline.

The Rockefeller shoeshine myth highlights an important but largely overlooked set of questions about the relationships between asset bubbles, the social composition of markets, and stratified outcomes in an era of renewed financialization (Kremp 2011). Recurring mass-participatory bubbles – in the stock market during the 1990s, in the housing market during the 2000s, and in cryptocurrency markets during the 2020s – beg the question of the degree to which differential market timing over boom-bust cycles is structured by social disparities in actors’ resources for navigating these high-uncertainty situations. In other words, to what extent do bubbles feed off social differentiation and inequality among market participants?

A sizeable behavioral-economic literature focuses on the social-psychological mechanisms that fuel upward price bidding, or “irrational exuberance” (e.g. Minsky 1986; Shiller 2005). However, behavioral models tend to treat each participant generically as the *n*th investor. Thus they have not grappled with the potential social-structural underpinnings of differential timing of

¹ Alternative versions of the myth feature either Joseph Kennedy or Bernard Baruch in lieu of Rockefeller, and some versions substitute a limousine driver as the proverbial working-class herald.

entry. Meanwhile, economic sociologists have shown how the segmented incorporation of households into an increasingly financialized economy fuels widening inequality (e.g. Lin and Neely 2020). However, sociologists have largely overlooked the *temporal* dimensions of asset market stratification.

This article proposes that in order to explain mass-participatory bubbles, it is necessary to consider *who* enters *when*. To do so, we build on the concept of *predatory inclusion*, a phenomenon whereby historically excluded groups later gain admittance into previously advantageous institutions, but on disadvantageous terms that subvert the presumed advantages of inclusion (Seamster and Charon-Chenier 2017; Taylor 2019). In the present context, *predatory inclusion* highlights the idea that bubbles depend on the presence of socially marginalized actors who can be more easily lured into the market during late stages of the boom, thereby sustaining continued upward price bidding even after incumbents become skeptical. The notion that differential timing of entry reflects differences in actors' social status is *implicit* in some classic historical writings on bubbles (MacKay 1841) but has not been tested directly (Kremp 2011).

However, it is not a foregone conclusion that asset bubbles will be sustained by incorporation of increasingly diverse and socially marginalized actors during later stages. While marginalization makes minority groups likely targets for financial exploitation, this same outsider status can also provide a buffer of protection by encouraging healthy distrust (Yenkey 2018). This is because diversity generates socio-informational frictions (Stark 2009), which discourage the micro-level processes of mimicry and contagion on which bubbles thrive (Levine et al. 2014). By contrast, homophily promotes interpersonal trust, undermining the vigilant skepticism that is needed to avoid falling prey to an emergent asset bubble (Yenkey 2018). As we elaborate below, this alternative buffering mechanism is not entirely at odds with the predatory inclusion thesis,

but it does deepen the question of whether and under what conditions we should expect social differentiation should help sustain bubbles through socially-segmented late-stage entry?

The present study tests and reconciles these alternatives by analyzing disparate timing of entry into the housing bubble in U.S. metropolitan neighborhoods during the mid-2000s. We focus on ethno-racial stratification, given the salience of race in structuring social boundaries and inequality in U.S. residential housing markets (e.g. Taylor 2019). Despite a large literature on the racialized nature of the mid-2000s boom and bust (e.g. Rugh and Massey 2010; Hernandez 2009; Faber 2013; Hwang et al 2015; Massey et al. 2016; Reid 2017), we know little about racial inequality in the *timing* of home purchases during this period. Prior research has focused on discrimination in lending terms within cohorts of buyers, rather than the extent to which actors are drawn into the market at differentially (dis)advantageous price points in the first place (conditional on lending terms).

We draw on tract-level housing price indices, coupled with Home Mortgage Disclosure Act data on mortgage-funded home purchases by race within tracts. We first test the overarching prediction derived from the predatory inclusion hypothesis by documenting racial differences in purchase timing over the local price cycle using semi-parametric generalized additive models. We find that home purchase rates for Black and Hispanic buyers diverged from those of whites as the bubble peaked in 2005, *even within the same mortgage market segments* (subprime/prime). Racial gaps in late-stage entry were especially pronounced in metropolitan areas with more acute overall price bubbles.

Building on the buffering perspective, we then consider how social influence and the structure of segregation within metropolitan areas may moderate differential demand responses to elevated prices. Consistent with theories of homophilous diffusion and network inequality (DiMaggio and Garip 2012), rates of late-stage subprime purchases by Black and Hispanic buyers

are highly responsive to lagged changes in same-race purchases in the tract. We also find that residential racial segregation amplified racial disparities in late-stage entry by rendering minority buyers less attuned to the emergent risks of purchasing a house at the bubble's peak, and more vulnerable to the overtures of predatory agents (Massey et al. 2016; Reid 2017). Taken together, our results support the micro-level mechanisms of homophilous contagion emphasized by Levine et al. (2014), but they point to different implications for the population-level relationship between race and bubble vulnerability. Within the segregated context of U.S. society, racial heterogeneity in market participation can sustain bubble growth and generate stratified exposure to financial risk, in part because diversification at the aggregate level often occurs within locally segregated interactional environments (Baker 1984).

The Social Dynamics of Asset Bubbles

Broadly, bubbles are defined as situations where the price of an asset rises rapidly before crashing, driven by social bidding dynamics divorced from fundamental forces of supply and demand (Kindleberger 1978, Shiller 2005). Asset bubbles have become increasingly frequent features of the U.S. and global economies since the 1980s. Compared to earlier eras, however, processes of financialization have rendered recent bubbles increasingly “mass-participatory” as credit market expansion and technological changes have enabled more non-professionals (including those with very small amounts of capital) to become entangled in volatile asset markets (Davis 2009; Goldstein and Knight 2023). For instance, whereas only about 3% of Americans owned stocks circa 1929, nearly 50% did by the peak of the “dot-com” stock market boom in 2001. Similarly, a boom in crypto-currency prices during 2020-2021 saw crypto-currency use extend to over 43% of men under age 35 in the U.S., including disproportionately large shares of ethno-racial minorities (Perrin 2021).

The Generic Stratification Structure of Asset Bubbles

Asset bubbles generate stratification based on order of entry (Shiller 2005). This structure can be conceived in relational terms: Early entrants' returns depend upon later entrants bidding prices higher, while later entrants' losses are due precisely to their attempts to mimic the successes of early entrants. Or as Keynes (1936) observed, "the actual object of the most skilled investment today is [...] to outwit the crowd, and to pass the bad or depreciating half-crown to the other fellow." Even those early entrants who still hold assets when the bubble bursts face lesser downside losses, having bought at lower prices than those who enter near the peak.

Figure 1 illustrates this stylized fact in the housing market context by showing the average cumulative asset appreciation circa March 2014, by year of purchase from 1997 to 2009. On average, houses bought in 2001 would have gained 10% in real value by 2014, while those bought at the peak in early 2006 would have lost over 30% in value eight years later. These differences are significantly more pronounced in regions with more pronounced price bubbles. Because they involve sharp price reversals, bubbles create conditions that amplify the significance of market timing.

[Figure 1 here]

Differential timing of entry is a functionally integral feature of bubbles because a continuing flow of buyers is required to sustain upward price bidding (Minsky 1986). Figure 2 illustrates this in the present case by plotting the Case-Shiller housing price index alongside the annual flow of owner-occupant home purchases (proxied by purchase mortgage originations). Real prices increased by 85% nationally from 1997-2006, significantly outpacing rental costs (as shown by the contemporaneous growth of the price-to-rent ratio). This means that even as prices rose relative to historical levels, buying also became less affordable from a consumption perspective.

Nonetheless, the annual flow of purchases grew by 72%. Note that purchases peaked alongside prices in 2005 and 2006. These buyers can be seen as late-stage entrants.

[Figure 2 here]

Behavioral Underpinnings of Heterogeneous Entry Behavior

Scholars have proposed a variety of social-psychological and institutional mechanisms to explain differential entry behavior during a bubble (see Turco and Zuckerman 2014). These models typically presume that, at a proximate level, varying responses reflect heterogeneous beliefs about future prices: As the bubble inflates, some actors recognize it as a bubble and cease to buy, while others view rising prices as a sign that prices will continue appreciating.

Indirect evidence of such heterogeneous expectations can be found in data on media coverage and internet searches. Contemporaneous discourses about bubbles index the extent of skeptical beliefs because classifying a market situation as a “bubble” indicates acknowledgment that price growth is tenuous and prone to reversal (Zuckerman 2010). Figure 3 shows that print media discussion and web searches about a housing bubble spiked in mid-2005, six months before the price peak, and two years before the larger financial crisis took hold (Case et al 2012). Although the presence of a nascent bubble is only ever fully clear in retrospect, a subset of actors often recognize the writing on the wall contemporaneously (Turco and Zuckerman 2014).

[Figure 3 here]

Evidence indicates, however, that late entrants remained optimistic about future housing price growth. Case and colleagues (2012) collected surveys of recent homebuyers from 2003-2012 to compare buyers’ future expectations against subsequent price dynamics. They find that, on average, those who entered near the market peak in 2005-2006 expected prices to

continue appreciating by 11% annually over the following ten years. Interview studies also reveal overriding optimism bias among buyers during the height of the housing boom (Reid 2010).

Bubbles and Social Differentiation: Predatory Inclusion or Protective Buffer?

The key issue for our analysis is the social-structural underpinnings of bubbles' temporally stratified role structure, particularly with respect to the racial stratification order that has long loomed large in U.S. housing and credit markets (e.g. Taylor 2019). How do social disparities shape the uneven timing of market entry over the course of the bubble's rise? And what can this tell us about the functional relationship between asset bubbles and social differentiation?

There is a longstanding intuition that differential timing of entry is driven partly by the fact that bubbles alter the relationship between social status and market participation. Charles MacKay ([1841] 2011), in his famous (though contested) account of the Dutch Tulip bubble, wrote of the successive process whereby first nobles and commercial classes, and then "farmers, mechanics, seamen, footmen, maidservants, even chimney sweeps and old clotheswomen dabbled in tulips."

Kindleberger (1978) also emphasizes how bubbles sustain themselves in part by widening the pool of participants: Rising prices and the accompanying social proof of quick riches entice more formerly inexperienced actors to enter (typically by means of credit). The implication is that compared to earlier stages, late-stage entrants are more likely to be lower-status and possess fewer socio-informational resources about the dynamics of the market. This expansion fuels the bubble by furnishing fresh recruits who bid up prices, but it also means that lower-status actors are systematically incorporated at less advantageous times. Insofar as bubbles entice those with the least market-relevant resources to enter later, they effectively feed off and reproduce existing forms of inequality. In this sense they can be seen as mechanisms of predatory inclusion.

In one of the only attempts to test this thesis empirically, Kremp (2011) assessed whether ownership of stocks diffused to underrepresented socio-demographic groups during the run-up to the 2001 “dotcom” crash. Although participation expanded among younger households, patterns along social status dimensions were less conclusive. Moreover, because this study used coarse triennial cross-sections, it was unable to capture finer-grained temporal stratification.

In partial contrast to the idea that bubbles feed off heterogeneity via predatory inclusion, another line of research emphasizes how social differences can inhibit bubbles (Levine et al. 2014), and exploitation in asset markets more generally (Yenkey 2018). Given that bubbles rely on mimicry and that actors are most likely to mimic similar others (Stark 2009), Levine and colleagues (2014) propose and find that greater ethno-racial diversity among market participants dampens upward price bidding by inducing decision frictions (relative to more ethnically homogenous markets). Similarly, Yenkey (2018) shows that ethnic outsiders are less susceptible to investment scams due to their distrust of dominant in-groups. This follows from the more general fact that predation typically relies on pre-existing trust, which tends to be stronger within networks and groups than between them (Portes 1998). Whereas the *predatory inclusion* perspective emphasizes the exploitative potential of social difference, the *protective buffer* perspective emphasizes the cautionary behavioral affordances of social difference that can help prevent marginal actors from succumbing to bubble traps.

We suggest that these perspectives can be reconciled by considering the distinction between heterogenous market participation and heterogenous market interaction. Since Baker’s (1984) early work on securities markets, economic sociologists have recognized that markets are socially segmented; actors rarely interact with the full market. Rather, information flows and exchange typically occur within smaller sub-networks. Thus, whereas diverse participation implies diverse interaction within the small world artificial markets studied by Levine and colleagues’ (2014),

high social segregation and homophily in real-world markets may foreclose such buffering: Even as market participation becomes more demographically diverse overall, information flows can remain locally homogenous.

Housing and Racial Inequality in the mid-2000s U.S. Housing Market

In this analysis, we focus on race, given its dual salience as an axis of both resource disparities and social group boundaries in the U.S. It is useful to provide some brief background on the institutional context of racial inequality in U.S. housing finance markets. The mid-2000s housing bubble occurred amid a longer-term shift in the dominant form of racial inequality, from discriminatory rationing of mortgage credit to racial segmentation within a dual mortgage market (Reid 2017). In this latter system, credit is more easily available to formerly excluded groups but often at higher cost and less favorable terms (Faber 2013).

Subprime lending was the primary engine of this shift. Subprime refers to higher-cost loans intended for borrowers with impaired credit. These products emerged during the 1990s before tripling in origination volume from 2003-2007. Subprime lending exhibited racial segmentation throughout. Numerous studies find racial minority borrowers are disproportionately likely to receive a subprime (rather than prime) loan, even after controlling for borrower and neighborhood characteristics. Some studies find that racial bias outweighs income, such that middle- and higher-income Blacks are statistically more likely than lower-income whites to receive a subprime loan (Faber 2013).

Considering predatory inclusion from the vantage of asset bubbles implies shifting analytical focus from credit terms to the timing of purchase. Analyses of mortgage defaults following the 2008 crisis suggest that timing of origination matters at least as much as mortgage

terms (Bayer, Ferreira, and Ross 2013), but studies of housing market stratification have largely overlooked variations in timing.

It is worth clarifying that there exist both long-term and short-term dimensions of racially stratified timing. Existing scholarship on disparate timing has focused on the longer-term. From this perspective, whites' greater access to mortgages throughout the post-war era allowed them to attain higher levels of homeownership during periods of stable growth. Meanwhile, the legacy of minorities' prior exclusion set the stage for their subsequent "hyper-inclusion" during the bubble (Hernandez 2009; Taylor 2019). This longer-term sequencing is an important part of the story, but, as we elaborate below, it cannot explain shorter-term temporal variations *within* the bubble period from 2000-2007.

Finally, it is important to highlight a few particularities of the housing market case. Unlike tulips, dot.com stocks, or other novel assets, the residential housing market was already institutionalized at the core of American society (e.g. McCabe and Rosen 2023). The strong cultural association of homeownership with upward mobility in the U.S. means that optimistic price expectations are underwritten by commonplace understandings of houses as "safe and solid" investments. These associations had been reinforced by decades of federal policies to encourage home-buying, including several homeownership promotion programs under the aegis of the George W. Bush Administration's "Ownership Society" agenda during the 2000s.² Moreover, housing's hybrid status as both investment and consumption implies that the behavioral processes through which actors are drawn into the market cannot be understood solely by reference to price expectations. Although the cost of homebuying relative to renting had also reached record levels at the bubble's peak (see the price-to-rent ratio series in Figure 2), symbolic and consumptive

² (<https://georgewbush-whitehouse.archives.gov/news/releases/2004/09/20040902-5.html>)

attractions of homeownership likely provided an additional source of fuel to sustain demand during later stages of the boom, an issue to which we return below.

Hypotheses: Race and Stratified Entry Timing

Our core hypothesis is that racial disparities in market-relevant resources and relational positions would leave members of marginalized groups more vulnerable to being enticed to buy at elevated prices on average, even *within* neighborhoods and mortgage market segments. Such predatory inclusion is expected to reflect the interaction of *both* supply- and demand-side mechanisms, and it implicates racial disparities in both market-relevant resources, as well as relational positions that make marginalized groups more attractive targets for predatory agents.

Differential Late-Stage Targeting: Race-Biased Mortgage Credit Supply

Heterogeneous timing of entry will evidence social stratification partly because those with fewer resources are more readily lured into the bubble by interested agents. Bubbles spawn an array of actors with an interest in attracting additional entrants (Kindleberger 1978; Minsky 1986). We refer to these actors as bubble agents. To the extent that bubbles are fueled by credit, lenders play a particularly important role in recruiting a continual supply of entrants. As discussed above, scholars have generally framed the racial bias of mortgage lending during the 2000s in terms of price discrimination among cohorts of applicants. Here we highlight how targeted credit also enticed more households to *enter the market at elevated prices in the first place*. Lenders, brokers, and agents do not simply function as gatekeepers who determine the terms under which borrowers can access credit; they can also serve as key agents in the stimulation of demand for homeownership (e.g. Besbris and Korver-Glenn 2023).

To understand why lenders were eager to recruit as many buyers as possible, it is important to recognize how the growing supply of mortgage credit after 2001 was fueled by the expansion

of securitization (Nadauld and Sherlund 2013). Originators faced especially intense pressure to maximize production of subprime mortgages because these served as the necessary building blocks for the most lucrative collateralized derivatives (Goldstein and Fligstein 2017). Given lower homeownership rates among Black and Hispanic populations, lenders could realize economies of scale by targeting communities with the greatest growth potential, who were also the most vulnerable to such exploitation (Rugh and Massey 2010; Hwang et al 2015).

Evidence indicates that rising late-stage entry rates among marginalized minorities reflected stimulation of demand as opposed to mere relaxation of borrowing constraints. First, adjusted denial rates for subprime mortgage applicants remained stable from 2002-2006 (Glaeser et al 2010, p.26). A constant denial rate implies that looser rationing of credit cannot explain the increasing number of minority entrants. Increasing originations had to reflect absolute growth in the number of applicants.

Second, qualitative data documents how lenders adopted concerted marketing strategies to entice racially marginalized buyers (Massey et al. 2016; Reid 2017). One practice involved recruiting trusted elites such as church ministers to market mortgages to parishioners. For each origination, the church would receive a donation from the bank (Massey et al. 2016). The greater prevalence of broker-originated loans within minority neighborhoods also reflected efforts to use ethnic embeddedness to facilitate sales. By enlisting co-ethnic brokers, originators could sell loans to borrowers who were distrustful of large banks (Steil et al. 2018).

Resource Disparities and Differential Susceptibility

Disparities in market-relevant resources also left Black and Hispanic buyers more vulnerable to late-stage entry. Asset bubbles can create perceptual traps insofar as rising prices reinforce the subjective view of an asset as a savvy investment, while quietly subverting the

objective foundations that had historically made that belief true: Americans who had been conditioned to view homeownership as a sure upward mobility path were thrust into a situation where market entry had become financially hazardous.

One key resource for navigating such minefields is the fluency and “know-how” gained from socialization and prior experience with real estate markets (Hayes 2020). Such tacit knowledge can help actors navigate uncertainty and judge others’ claims-making about the future, making potential purchasers less easily swayed by the breathless promises of unscrupulous lenders. Decades of greater exclusion from housing and investment markets had left Black and Hispanic actors with systematically less prior experience. This will heighten their susceptibility to late-stage entry, especially in a context where predatory lenders sought to exploit information asymmetries with inexperienced buyers.

Racial disparities in susceptibility to the bubble trap may also follow from cultural frames about homeownership and mobility (McCabe 2018). Homeownership *per se* tends to be viewed as a more meaningful achievement among historically excluded groups because it is associated with upward mobility and “beating the odds” (Greif 2014). For whites, who are almost twice as likely to have grown up in a home-owning household, achieving homeownership may evoke less acute symbolic associations (Greif 2014). The more idealized homeownership as a mobility strategy, the greater the cognitive leap required to recognize potential hazards. The taken-for-grantedness of upward mobility frames also makes it easier for bubble agents to exploit such perceptions.

Taken together, the above arguments imply that accumulated racial disparities in market-relevant know-how left Black and Hispanic actors more attractive targets for predatory agents, less well-equipped to navigate the claims of those agents, and thus more vulnerable to being enticed to buy at higher price points on average. Given lenders’ greater incentives to exploit

systemic inequality when recruiting buyers with higher-cost mortgages, we expect that temporal stratification will be particularly pronounced among buyers who enter via subprime financing. However, a key point for our argument is that inter-group differences in timing of entry *cannot* be reduced solely to the racially-targeted expansion of subprime lending during the late stages of the boom. Notably, most purchase mortgages to Black and Hispanic borrowers during the peak bubble years from 2004-2006 were not subprime (based on the HMDA lender indicator). This fact highlights the significant concurrent role of conventional lending in late-stage entry, and it raises the possibility of a more general temporal dimension of racialized predatory inclusion that occurs *within both* the prime and subprime segments of the mortgage market.

We thus hypothesize that even when stratifying by mortgage type and controlling for variations in local credit availability, existing race-specific homeownership rates, and other concurrent features of the local housing market (via fixed effects), Black and Hispanic purchases peaked later in the local price cycle compared to white purchases.

Hypothesis 1a: Conditional on local credit availability, race-specific homeownership saturation, and mortgage type, Black and Hispanic buyers will tend to be incorporated into the market at higher price points compared to white buyers.

Hypothesis 1b: Racial disparities in late-stage entry will be relatively greater in the subprime mortgage market segment, where bubble agents have greater incentives to take advantage of systemic racial inequality.

Hypothesis 1 above focuses on conditional *average* racial disparities in timing of entry. However, such temporal stratification may be particularly pronounced in social settings that concentrate vulnerability.

We expect racially disparate responses to rising prices to be wider in cities where steeper price booms accentuated the perceptual trap of late-stage entry. At their peaks, the Phoenix and Las Vegas MSAs saw annual appreciation above 45%. Meanwhile, other cities – including growing ones like Atlanta, Charlotte, and Dallas – experienced more modest annual appreciation

of approximately 10%. Where price bubbles are more acute, the demand-side racial differences in market experience and information we outlined above become more salient, and thus the opportunities for supply-side actors to exploit such vulnerabilities increase. In other words, if bubbles generally tend to promote predatory inclusion, then places with more acute bubbles should exhibit more pronounced racial stratification in timing of entry.³

Hypothesis 2: Between-race differences in average demand responses to housing price growth will be wider in metropolitan regions where the amplitude of the price bubble was more pronounced.

Homophily, Network Inequality, and Spatial Segregation

Finally, two other factors that may heighten susceptibility to the bubble trap concern informational segregation in the local environment (Baker 1984; Levine et al. 2014). Actors formulate strategies based partly on cues from others. Indeed, it is only through social ties and signals that economic agents can plausibly devise courses of action under conditions of uncertainty (Beckert 1996). However, actors' differentiated positions in social and geographic space expose them to different signals: the stories and advice one hears, and social proof one observes. Disparate network resources and segregation thereby furnish potential buyers with unequally useful information, which in turn affects their ability to navigate uncertainty (DiMaggio and Garip 2012).

Local contextual cues and network inequalities could contribute to racial disparities in late-stage entry through a mechanism of homophilous diffusion. This refers to the fact that actors are more likely to adopt or avoid a practice when socially similar others have done so. The tendency to mimic similar others is strongest when they are geographically proximate (McPherson et al 2001). This is especially salient for housing because the market is subjectively

³ In a supplementary analysis reported in the Appendix, we also consider whether racial disparities in timing might differ in more and less supply-constrained areas.

understood to be localized. The social proof of others' successes – whether direct network ties or socially similar strangers – also reinforces the credibility of bubble agents' marketing claims.

Homophilous diffusion implies that small initial differences in behavioral responses to the bubble will tend to propagate within groups: When well-informed actors shift beliefs or behavior based on new information, homophily ensures that socially similar others receive the signal before dissimilar others. Although all actors are prone to homophilous cues, the mechanism has uneven implications across groups due to uneven stocks of resources (Zhao and Garip 2021). In other words, homophily compounds the effects of inter-group differences in the informational resources of individual members (DiMaggio and Garip 2012).

A second parallel mechanism is segregation (Faber and Drummond 2024). Residential segregation helps perpetuate inequality by insulating marginalized groups from access to advantageous information (Krysan and Crowder 2017; Besbris, Schachter, and Kuk 2021). To the extent that neighborhood segregation reduces actors' exposure to countervailing signals in local housing markets, we would expect segregation to amplify racial differences in demand responsiveness to price changes (Levine et al 2014). In other words, segregation could contribute to racial disparities in late-stage entry not only by presenting lenders with pockets of exploitable demand (Hwang et al. 2015), but also by concentrating the perceptual feedback effects of social cues and thereby heightening vulnerability to the bubble trap.

Prior qualitative research offers compelling evidence on the relevance of these mechanisms. For instance, Reid's (2010) interview study of lower-income homebuyers during the late stages of the bubble found that ties to other recent buyers served as powerful sources of *both* motivation *and* information. Buyers often became convinced to buy after observing the successes of friends and relatives. Network ties actively encouraged buying and assuaged concerns. Networks also helped propagate late-entry through referrals to bubble agents by imprinting third-party claims

with social trust (Uzzi 1996). Reid's (2010) snowball sample uncovered five separate underwater borrowers on a *single block*, all of whom had acquired sub-prime mortgages from a single mortgage broker to whom they had referred one another. Such dynamics highlight how supply-side targeting interacts with network inequality and segregation among potential buyers.

The above arguments suggest that stronger homophilous cues and greater racial residential segregation will each exacerbate between-group disparities in late-stage entry. In other words, the protective buffer mechanism will be less operative in contexts which are more internally segmented and same-group signals are stronger because such segmentation isolates marginalized groups from countervailing signals. These factors are also key to reconciling the *predatory inclusion* and *protective buffers of diversity* perspectives: To the extent that overall diversification of market participation occurs through locally segregated and unequally-resourced sub-networks of information flow (DiMaggio and Garip 2012; Zhao and Garip 2021), the protective benefits of market diversity identified by Levine et al. (2014) will remain inoperable: There is diverse participation in the aggregate, but not diverse local interaction.

As above, the effects of homophilous cues and segregation are expected to be especially pronounced where bubble agents have stronger incentives to more fully exploit the informational inequalities to which these factors give rise (i.e., within the subprime lending segment).

Hypothesis 3a: Stronger homophilous cues within a market will increase the subsequent association between rising prices and entry rates among buyers from more marginalized groups, but not among buyers from less marginalized groups.

Hypothesis 3b: Racial disparities in the effects of homophilous cues will be relatively greater in the subprime segment, where lenders have greater incentives to take advantage of systemic racial inequality.

Hypothesis 4a: Greater metropolitan-level residential racial segregation will be associated with a more strongly positive association between locally rising prices and entry rates among Black and Hispanic buyers.

Hypothesis 4b: Racial disparities in the effects of segregation will be relatively greater in the subprime segment, where lenders have greater incentives to take advantage of marginalized groups' spatial isolation.

Data and Methods

We analyze locally aggregated tract-level administrative data from the Home Mortgage Disclosure Act files (Pettit 2013), which cover most mortgage-funded purchases in non-rural areas. HMDA is the only comprehensive national data source on home purchases by race. It is widely used in studies of mortgage market stratification. We merge HMDA data on race-specific home purchase counts in census tract-years (Pettit 2013) with annual tract-level housing price indices to gauge time-varying purchases by race relative to local prices cycles (Bogin, Doerner, and Larson 2019). Our analysis covers the period from 2000-2006. More detail on the data sample is presented in the supplementary appendix.

Outcome and Covariates

The primary outcome of interest is the number of mortgage-funded purchases by race in a tract-year. Specifically, we define *market entry* as the purchase of an owner-occupied residence using a mortgage, which we operationalize using first-lien purchase (not refinance) originations. Restricting the outcome metric to mortgage-funded purchases excludes most “downsizing” moves, in which buyers do not enter the housing bubble in the same way as those who make a debt-funded investment.⁴ Our focus on timing of entry (rather than entry and exit) reflects housing’s shelter-use and its relative illiquidity. Strategically timed exit is less relevant for owner-occupied housing compared to other assets, given the difficulty of selling quickly, and the transaction costs of securing alternative housing.

⁴ The proportion of all-cash U.S. home sales hovered around 20% from 2000-2007.

The ethno-racial groups included in the analysis are non-Hispanic white, Black, and Latino/Hispanic, which are originally reported in mortgage application forms and then submitted by lenders to HMDA. We exclude Asian, Pacific Islander, Indigenous and other non-white groups from the sample both due to small sample sizes and because their position in the housing market is less well-documented in prior literature, making comparisons using these groups less clear as tests of our theorized mechanisms.

We measure housing price appreciation at the tract-level, using an index that is locally normalized to equal 100 in the year 2000. Tract-level HPI data come from an experimental series constructed by economists at the Federal Housing Finance Authority (Bogin et al. 2019). They show that over a quarter of the total variance in neighborhood price growth from 2000-2005 was within rather than between cities. Measuring “time” in the boom-bust cycle at the tract-level effectively adjusts for varying amplitude and timing of price peaks across locales. The index is “constant-quality”, meaning that it is adjusted to account for changes over time in housing size composition within a tract. To test Hypothesis 2, we create strata of local bubble intensity using the difference between the HPI peak and the post-crash price trough. We group tracts into three bubble intensity strata: 1st-60th percentile, 61st-80th percentile, and 81st-99th percentile. We also tested the robustness of our findings to alternative house price normalizations (see Appendix).

Mortgage type (“prime” versus “subprime”) is distinguished using the pre-2004 U.S. Dept. of Housing and Urban Development’s list of high-cost, or primarily nonprime mortgage originators (Petit and Droesch 2008). This lender-based indicator may undercount subprime mortgages during the later years of the bubble because it does not include those high-cost loans made by lenders that primarily originate prime-rate loans. We use the lender-based indicator to maintain measurement consistency over the study period, but we subjected this approach to

sensitivity tests by substituting HMDA's more precise rate-spread indicator starting in 2004.

Results are substantively identical (see Appendix).

We control for over-time variations in credit rationing or availability using annual tract-level, race-specific mortgage denial rates from HMDA. We include both the lagged and current year denial rate for all mortgages.

Finally, white buyers could become scarcer during late stages of the boom simply because white homeownership had reached a level of near saturation in some areas, while the pool of potential first-time Black and Hispanic homebuyers remained relatively larger given their lower baseline homeownership rates. To address this, we also control for metropolitan-level racial population shares and race-specific homeownership rates. Together, these adjust for variation in the pool of potential buyers. These measures are based on CBSA-level Census figures, which we interpolated annually between the 2000 Decennial and 2007 3-year ACS.

To test Hypotheses 3a and 3b (homophilous cues), we construct a tract-level measure of same-race entry using the prior year's race-specific rate of purchase mortgage originations. Specifically, we calculate the lagged, tract-standardized count of HMDA race-specific purchase mortgages in the tract where an individual buys in the focal year. Importantly, this measure does not reflect exposure to same-race buyers' participation in the housing boom in buyers' previous residential locations. Rather, it captures recent increases in same-race buyers' purchase activity in the *destination* neighborhood. This aligns with our theoretical mechanism, which posits that homophilous cues (e.g. information about specific neighborhoods) heighten buyers' responsiveness to localized bubble dynamics.⁵ To test Hypotheses 4a and 4b, we measured racial

⁵ We examined several alternative operationalizations of recent same-race market entry, including lagged absolute purchase *growth*, lagged percentage change in same-race purchases, and lagged unstandardized number of same-race purchases. All results yielded substantively the same conclusions as our main analysis.

segregation at the metropolitan level using the dissimilarity index between each respective minority group and whites.

Model Estimation

Because local housing price trajectories and race-specific entry rates are expected to exhibit a nonlinear relationship of unknown form, we estimated regression-adjusted trends in the number of purchases by racial group and mortgage type over the price cycle using a semi-parametric Generalized Additive Model (GAM) approach (Wood 2017). For each respective racial group and mortgage financing type, we estimated a separate model of the form,

$$N_{ijt} = f(HPI_{it}) + \gamma X_{ijt} + v_i + \epsilon_{it} \quad (1)$$

, where the number of purchases in a tract i in metropolitan area j in year t is a nonlinear smoothed function of locally normalized housing prices (tract HPI), which is estimated from the data using a penalized regression spline; the linear control variables discussed above, and tract fixed effects. Inclusion of tract-fixed effects controls for the uneven racial population distribution across locales, as well as any other time-invariant confounding features of the neighborhood.

The key quantity of interest for assessing Hypotheses 1a and 1b is the regression-adjusted timing of peak market entry rates across locally-normalized price curves. This quantity is ascertained by plotting the estimated local polynomial function for the partial regression component f across values of the tract-level housing price index value for each racial group. Models are run separately for prime and subprime market segments in order to assess our hypotheses of stronger effects within the subprime segment.

To test Hypothesis 2, we augmented the basic model above by allowing the semiparametric smoothed local housing price term to vary by bubble-intensity strata. The model, again estimated for each racial group and mortgage type, takes the form,

$$N_{ijts} = f_s(HPI_{it}) \cdot s_i + \gamma X_{ijt} + v_i + \epsilon_{it} \quad (2)$$

where s is a vector of dummy variables indicating the bubble intensity strata of tract i and f is a vector of semiparametric smoothed functions corresponding to each stratum.

To test the moderation effects in Hypotheses 3-4, we add to Equation (1) additional smoothed interaction terms between the tract HPI and each of the respective moderators: homophilous diffusion and metropolitan segregation. In the homophilous diffusion case, the model takes the form,

$$N_{ijt} = f(HPI_{it}) + N_{ij,t-1} + f(HPI_{it}, N_{ij,t-1}) + \gamma X_{ijt} + v_i + \epsilon_{it} \quad (3)$$

where both the present and lagged number of purchases are standardized within tracts over time. The model is similar for the segregation moderation analysis, with metro-level racial segregation replacing the lagged number of purchases.⁶ These models are again run separately for prime and subprime market segments.

We test Hypothesis 3 by plotting point estimates of the first derivatives of the smoothed HPI function across the distribution of lagged same-race purchases. The quantity of interest here is whether the difference between white and Black or Hispanic responses to price increases are *greater* when same-race purchases were higher in the previous year. We test Hypothesis 4 by plotting the predicted number of race-specific purchases across HPI levels separately for tracts in MSAs at the 20th and 80th percentile respectively of the metropolitan segregation distribution. The relevant quantity of interest here is whether entry rates for Black and Hispanic buyers are more strongly associated with rising prices in higher-segregation metros than in low-segregation metros.

⁶ In the segregation moderation analysis, we use metropolitan area fixed effects instead of tract fixed effects because racial segregation is measured at the metropolitan level and does not vary at any sub-metropolitan level.

Results

National Descriptive Patterns of Racially Disparate Timing

We begin by documenting aggregate racial differences in the timing of market entry. Figure 4 depicts inter-group differences using three different metrics. All three show a common trend in which purchases by Blacks and Hispanics increased as the bubble neared its peak, while whites' purchases diminished. The left panel shows the annual year-over-year percentage growth in race-specific rates of entry. Before 2003, all three groups display a similar cyclical pattern, though whites appear to lead the other two groups by one year. After 2003, rates diverge, with white buyers exhibiting relatively stable 7-9% annual growth. In contrast, entry by minorities *accelerated* in 2004. As the bubble was peaking nationally, Black and Hispanic entrance rates were still growing at an annual rate of 25-30%..

[Figure 4 here]

A second measure of temporal disparities is the *over-/under-representation* of buyers over the price cycle. The middle panel of Figure 4 shows each group's annual share of all owner-occupant purchase mortgages, expressed as a ratio of that group's share of household heads. A value of 1 indicates that a group's representation among new entrants is proportionate to its share of the population. Hispanic households went from being underrepresented during the early years of the bubble to dramatically over-represented as prices peaked. Although Black buyers remained underrepresented throughout the period, they also exhibited a pronounced upward trend during the last years of the boom. In contrast, whites' relative representation declined after 2003. By the market peak in 2005-2006, whites were underrepresented among U.S. homebuyers for the first time in history.

Finally, the right panel of Figure 4 shows racial differences in the within-group temporal distribution of purchases: Of all mortgage-funded purchases made by buyers of a given group from 1999-2006, what proportion of purchases occurred during the peak years of 2005-2006? Here again the HMDA data tell a consistent story. Among whites, 28% of purchases occurred during 2005-2006. For Blacks, that figure rises to 36%. And for Hispanics, it rises to 40%.

Conditional Racial Differences in Timing of Entry (Hypothesis 1)

The aggregate national results reported in Figure 4 do not account for the uneven distribution of groups across regions with varying intensity and timing of bubbles. Nor do they distinguish prime and subprime financing channels.

Figure 5 shows covariate-adjusted GAM model estimates, with “time” (the x-axis) represented by tract-specific housing prices relative to year 2000. In this and the following figure, the Y-axis represents the standardized number of purchases made in a tract-year by each respective racial group (expressed in standard deviation units). Across the prime and subprime segments, we consistently see that even when controlling for race-specific homeownership saturation, and time-varying credit constraints, purchases by Blacks and especially Hispanics peak at higher price points compared to whites. Among buyers with prime mortgages, whites (blue line) reached their peak purchase rates—about 0.15 standard deviations above their mean—when house prices were 25 HPI points above the tract average. Black buyers (red line), by contrast, peaked later in the bubble, when prices were 50 HPI points above the tract mean. Hispanics’ (green line) purchase rates never clearly peak, remaining about 0.5 standard deviations above their mean even at the highest price levels before the crash. Broadly similar patterns appear among buyers with subprime loans.

Together, these results are consistent with the prediction of Hypothesis 1a: conditional on mortgage type, credit availability, lagged race-specific homeownership saturation, and tract-fixed effects, Black and Hispanic buyers were drawn into the market at less advantageous stages of the local price cycle. In other words, regardless of where they bought and with what type of mortgage, Black and Hispanic homebuyers from 2000-2006 were more likely to do so when prices were already high. By contrast, even white buyers who enter the market via subprime financing tend to do so at more advantageous points in the market cycle.

Hypothesis 1b finds only partial support in Figure 5 insofar as racial disparities in timing of entry are *not* consistently more pronounced in the subprime segment compared to the prime segment across all groups. The temporal gap between Black and white buyers in the extent of late-stage purchases is relatively more pronounced in the subprime market, as indicated by the distance between the peaks of the red and blue curves, but the inverse is true for Hispanic buyers.

[Figure 5 here]

Local Context: Magnitude of Local Price Bubble (Hypothesis 2)

Hypothesis 2 predicted that there would be wider racial disparities in purchase rate response to rising neighborhood prices in metropolitan areas with more pronounced overall bubbles. Figure 6 shows the results of this analysis, again controlling for the same covariates as above. The plotted estimates show the same quantities as in Figure 5, but separately across three different strata of local cumulative peak appreciation.⁷

Here we see that the between-race disparities in marginal behavioral responses to rising prices grow wider in cities where the overall bubble was more acute. The higher the peak price point, the greater the degree of within-city racial disparity in purchase rate response at the height

⁷ Results are substantively identical when we measure bubble intensity as the difference between the peak and the price level in 2000, and as the average of the two former differences.

of the local boom. For example, when prices are 50 HPI points above their tract mean in places with the most intense bubbles, the Black-white gap in subprime purchases is 0.5 standard deviations, whereas in places with bubble intensities in the lowest 60%, this gap is statistically undetectable. This pattern supports Hypothesis 2.

[Figure 6 here]

Local Context: Homophilous Diffusion and Segregation (Hypotheses 3a-4b)

Hypothesis 3a (homophilous diffusion) predicted that recent growth in same-group entry rate will heighten the association between prices and willingness to buy at elevated prices to a greater degree among Blacks and Hispanics compared to whites. Hypothesis 3b further predicted this disparate pattern will be stronger in the subprime segment.

To test these arguments, we modeled the race-specific market share using an interaction between the lagged same-group market share and tract-level prices, net of the same controls as above. Figure 7 plots the estimated average marginal effects of local prices on entry rates (i.e., the race-specific average elasticity of price response) across the distribution of same-group market-share in the previous year, again centered at the tract-specific mean. When interpreting the plots, it is important to bear in mind that any Y-axis values above zero represent a positive housing demand response to rising prices and that X-axis values above 0 indicate above average same-race purchases in the same tract in the previous year.

The results in Figure 7 lend partial support to the predictions of H3, but only within the subprime market segment (which is consistent with H3b but not H3a). Among those purchasing with conventional mortgages, there is minimal evidence of racially disparate social influence on price responsiveness (Figure 7 left panel). Black buyers' price elasticity increases with previous-year market entry rates, which we would expect based on the theory. But so too does

white buyers' price elasticity (though not statistically above zero). Moreover, Hispanic buyers' elasticity actually falls at high levels of same-group lagged entry.

There is clearer evidence of disparate effects of homophilous cues within the subprime market segment (H3b). Here we see increasingly positive price elasticity for *both* Black and Hispanic buyers (but not white buyers) as a function of same-group lagged growth in market entry rates. This latter result is consistent with H3b, which predicted that the stronger incentives of bubble agents working in the subprime segment to exploit buyers' in-group social trust (Reid 2010; Steil et al. 2018) would result in a more disparate effect of homophilous diffusion on late-stage entry.

Taken together, the results in Figure 7 support the argument that the behavioral signal of same-race buying is associated with heightened willingness to buy even at highly elevated prices (Levine et al. 2014). However, only within the subprime market segment does this micro-level mechanism help explain the racially disparate late-stage entry.

[Figure 7 here]

Hypotheses 4a and 4b predicted similarly that racial segregation would heighten the association between price growth and Black and Hispanic market entry. Figure 8 shows how estimated purchase rates for Black and Hispanic buyers vary across the tract-level price curve, contrasting otherwise comparable tracts in MSAs with low (20th percentile) versus high (80th percentile) levels of metropolitan segregation. These are derived from models that non-parametrically interact metropolitan-level segregation with tract-level prices. For example, at a given level of price appreciation (X-axis), greater estimated Black purchase rates in tracts located in MSAs at the 80th segregation percentile versus the 20th percentile indicate that Black buyers in more segregated areas respond more strongly to rising prices than in less segregated areas.

As with homophilous diffusion, these results reveal that racial segregation accentuated marginal groups' susceptibility to the bubble trap only within the subprime market segment – not within the prime segment. This is consistent with H4b, but not H4a. Black and Hispanic buyers using subprime loans exhibit stronger demand in the face of elevated neighborhood prices in more segregated cities compared to same-group responses in less segregated cities). The fact that that this mechanism was confined to subprime buyers suggests that Black and Hispanics' geographic isolation only translated into disproportionate late-stage entry to the degree that lenders also had strong incentives to take advantage of the information asymmetries generated by that isolation (cf. Rugh and Massey 2010). Despite this important qualification, Figure 8 provides some support for the theory that Black and Hispanic homebuyers were more susceptible to the late-stage bubble trap in more segregated cities.

[Figure 8 here]

Discussion and Conclusion

The deepening integration of households into the financial economy over the past two decades makes asset markets key sites to examine emergent mechanisms of social stratification (Lin and Neely 2020). The present study uses the case of the mid-2000s U.S. housing market to assess racial stratification in timing of market entry across boom-bust price cycles.

We find that conditional on financing terms, credit availability, and local race-specific homeownership saturation, trends in purchase rates for Black and Hispanic buyers diverged from those of whites as the bubble neared its peak in 2005. We further find partial evidence that residential segregation and processes of homophilous diffusion exacerbate these racial disparities in late-stage entry by rendering actors less attuned to the emergent risks of purchasing a house at the bubble's peak, and more vulnerable to the overtures of predatory agents.

These findings shed new light on the social underpinnings of mass-participatory bubbles, and racial stratification in the mid-2000s housing boom in particular. By shifting analytic focus to disparities in timing of purchase over the market cycle, our analysis foregrounds a dimension of inequality – *timing* – which has been largely overlooked in the literature on racial inequality in homeownership markets (e.g. McCabe and Rosen 2023). Black and Hispanic buyers not only tended to receive credit on less advantageous terms; they were also drawn into the market at less advantageous *price points* relative to white buyers.

The analysis also helps to reconcile two different views about the relationship between social differentiation and vulnerability to bubble traps. The temporal stratification by race that we document above may seem at odds with the findings that diversity in asset markets provides cautionary affordances that dampen the mimicry on which bubbles feed (Levine et al. 2014), and buffer minorities from predation (Yenkey 2018). These perspectives can be partly squared, however, by considering the role of residential segregation and homophily. We showed that among subprime buyers, Black and Hispanic buyers were more likely to succumb to the bubble trap (i.e. exhibit more positive demand response to price growth) when their same-group *local* market share grew in the previous year (Fig. 7). We also showed that racial minorities buy at greater rates during late stages in more residentially segregated neighborhoods (Fig. 8). We explain these patterns by suggesting that home purchase decisions are shaped by social cues, via both information diffusion in homophilous social networks and informational isolation in segregated contexts.

Even if the overall racial composition of the housing market became more diverse over time, the segmented structure in which it grew prevented the mechanisms which Levine et al. identify from taking hold (Baker 1984). The micro-level frictions that they find to curb bubble behaviors are only possible in a social context where racial groups are allowed to interact more or

less freely. In the housing markets of the mid-2000s, however, potential late-stage buyers were insulated to a significant degree from dissonant signals (Stark 2009) due to social and geographic segregation. Social differentiation can help fuel bubbles in the real world precisely because heterogeneous actors occupy stratified networks. A larger implication is that the degree to which diversification of an asset market fuels or impedes bubble formation depends not only on the racial composition of the market, but also on its internal relational structure and the opportunities that structure provides for intergroup interaction.

The analysis also makes several broader contributions. First, we bring a distinctly sociological perspective to the literature on asset bubbles (Shiller 2005) by attending to the role of social differentiation in sustaining them. Market participants are not generic investors. Rather, they are situated within stratified structures which furnish them with uneven resources for navigating uncertainty, and uneven vulnerabilities to predation. To the extent that continued price-bidding is sustained partly by downward social diffusion of participation, inequality among market actors should be seen as integral rather than incidental to how a bubble unfolds.

Second, the analysis draws new linkages between research on financialization and stratification. Economic sociologists have documented how the uneven incorporation of households into the financial economy affects distributional patterns of income and wealth (e.g. Lin and Neely 2020), but we know less about how actors' resources and relational positions shape their unequal chances (i.e., returns) *within* asset markets. We link these traditions by illustrating how mass-participatory asset bubbles – a key outgrowth of financialization – are structured by and reinforce durable forms of inequality (Tilly 1998).

Here, our findings add to literature on racialized predatory inclusion processes (e.g. Seamster and Charon-Chenier 2017). Because they generate powerful perceptions of opportunity in an environment marked by uncertainty and incentives for malfeasance, asset bubbles are

particularly conducive to predatory inclusion. Our finding of amplified vulnerability to late-stage entry among Black subprime buyers in more segregated cities also buttresses claims about how racial segregation enables market predation (Faber and Drummund 2024). Of course, characterizing asset bubbles as mechanisms of predatory inclusion should not imply a crude determinism: All market actors face uncertainty, and in this case millions of white homebuyers also succumbed to the bubble trap. The theory is rather a probabilistic one, whereby actors in more marginal positions tend to be incorporated at what turn out to be less advantageous moments in the price cycle.

Future research might extend and refine this model by analyzing dynamics of asset market stratification across multiple dimensions of inequality. In the present context, race operates both as a categorical basis of predation and as a salient axis of resource disparities. Estimating stratified timing by social class status within racial groups could help disentangle the relative contributions of these mechanisms.

Future research might also test the reach of this explanatory framework to other assets and contexts. For instance, do analogous patterns of inter-group stratification appear in other countries' contemporaneous housing bubbles during the mid-2000s, such as Spain? And do similar temporal disparities appear in emergent asset markets such as crypto-currencies, which are also characterized by diverse participation, predatory recruitment practices, and upward mobility/opportunity discourses (Olson and Lewis 2023), but which lack housing's institutionalized financing system and dual status as shelter and investment? Our framework predicts the answer in both cases would be yes, but this is an open empirical question.

Finally, attending to the social topology of asset price cycles highlights the need for a more dynamic understanding of racial wealth stratification (Flippen 2004; LaBriola 2024). Advocates and government policies have long sought to mitigate wealth gaps by promoting

investment in historically high-appreciation assets such as stocks and housing (Denton 2001). However, the serial bursting of asset bubbles in recent decades highlights how inclusion also brings risks of disparate wealth destruction by exposing late-entering households to sizeable losses. Whereas research on wealth gap trajectories since the 2007 crash has focused on geographic exposure, portfolio composition, foreclosure, and disparate neighborhood appreciation rates (e.g. Pfeffer, Danziger, and Schoeni 2013), our results suggest scholars should also pay greater attention to stratified *temporal* processes of incorporation in volatile asset markets.

Data Availability Statement

The data and code to replicate analyses reported in this article are available in the Open Science Foundation (OSF) repository, accessible at https://osf.io/8y4xw/?view_only=e9985aac4c324d488715447bfa1fc36e >

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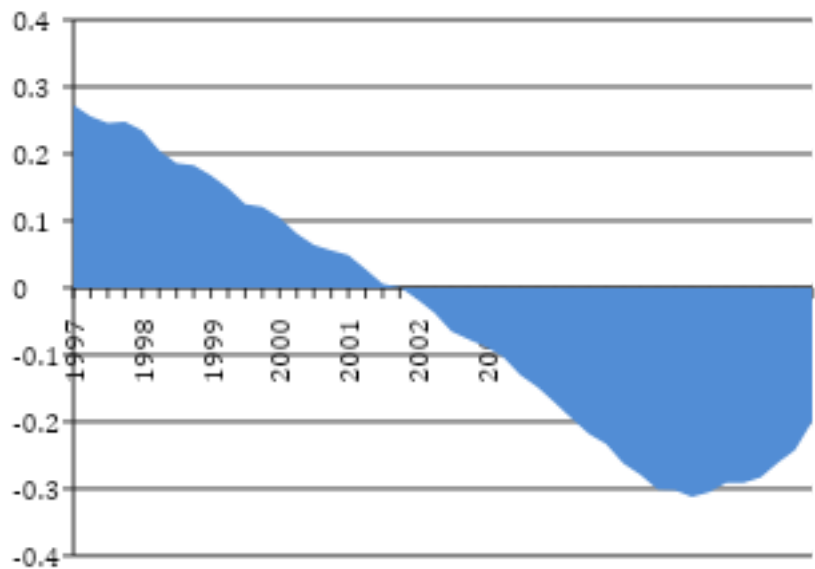
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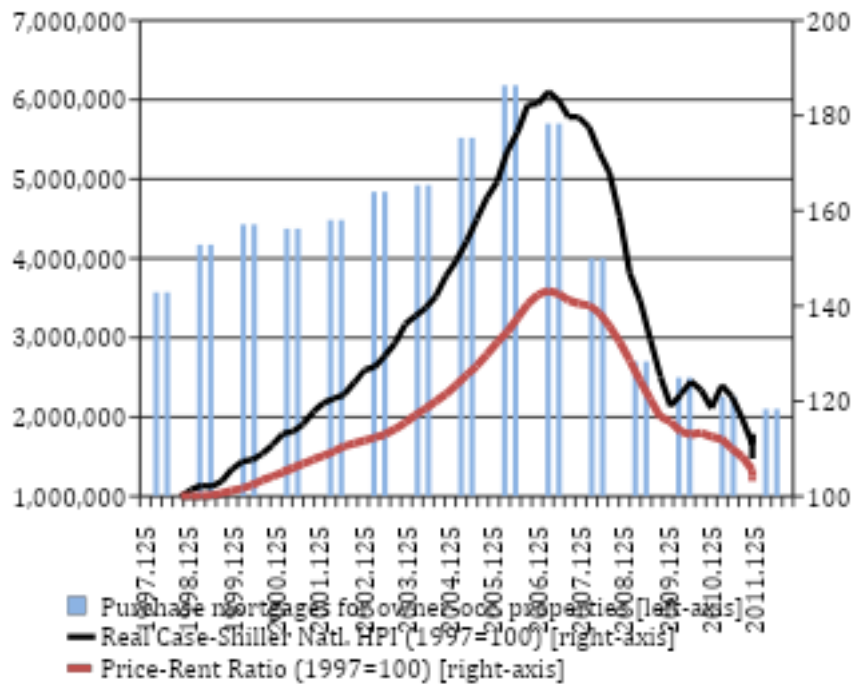
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Figure 1: Average Expected Cumulative House Price Appreciation from Time of Purchase through March 2014, by Time of Initial Purchase (based on national HPI index)



Data Source: Case-Shiller National Quarterly Housing Price Index

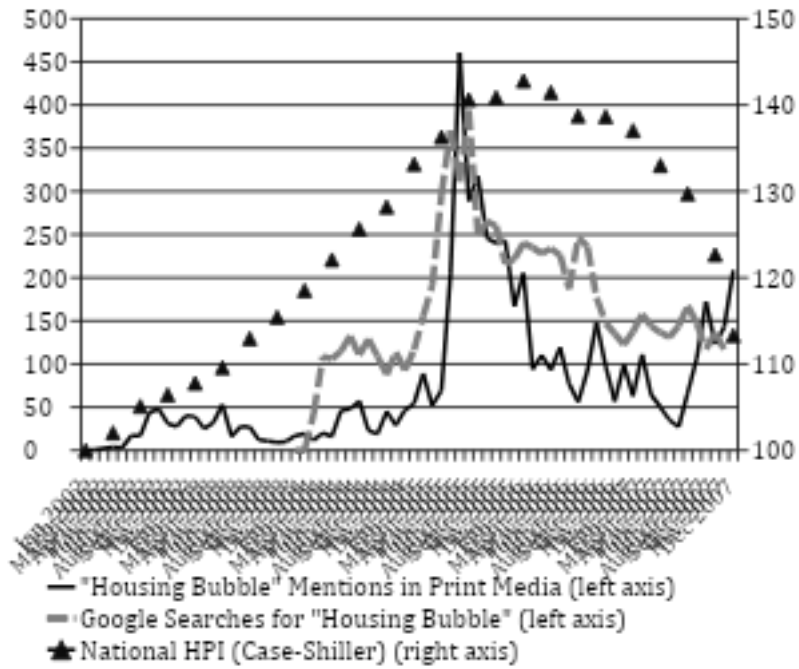
Figure 2: Housing Price Appreciation and Annual Flow of Owner-Occupant Purchase Mortgage Originations



Data Sources: Case-Shiller National Quarterly Housing Price Index and HMDA.

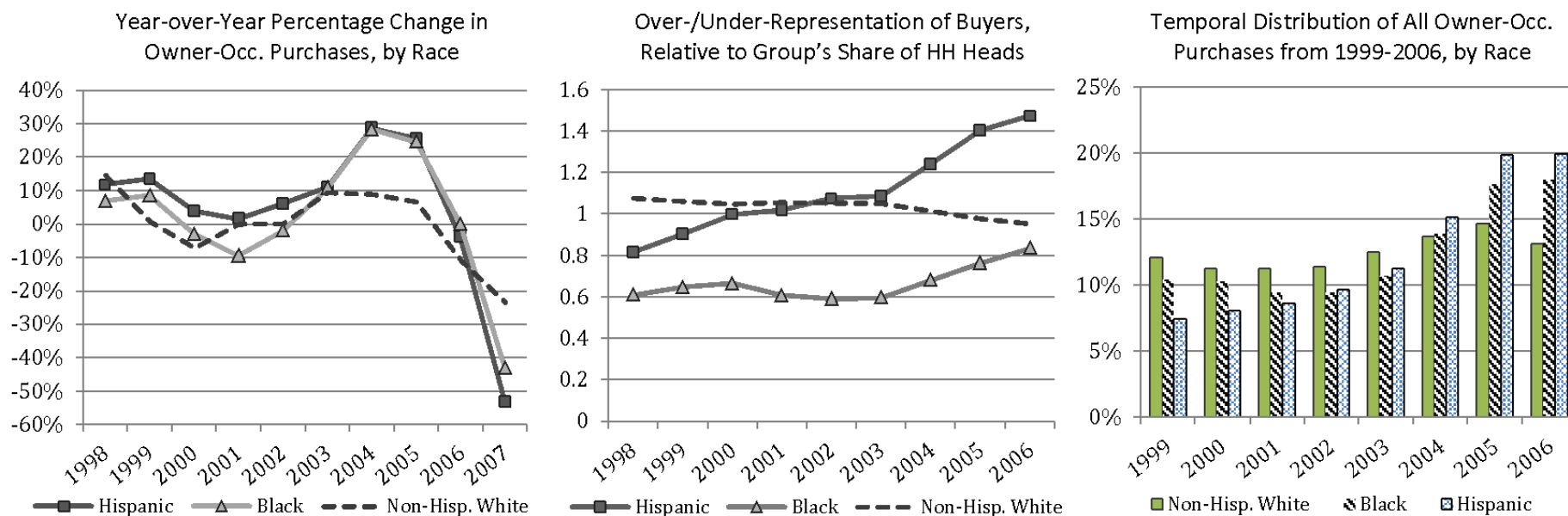
Note: Figure 2 shows the Case-Shiller national housing price index (inflation-adjusted and indexed to 1949=100), and the annual number of HMDA reported owner-occupant mortgage for purchases of 1-4 family unit housing.

Figure 3: Media Mentions of “housing bubble” and Google Searches Nationally



Note: Figure 3 gauges contemporaneous awareness about the housing bubble based on keyword content analyses in national and local media outlets. We tabulated the number of articles *and* cable television news segment transcripts which mentioned the terms “housing bubble”, “house price bubble”, or “real estate bubble” using the Lexis-Nexis and NewsBank Access databases.

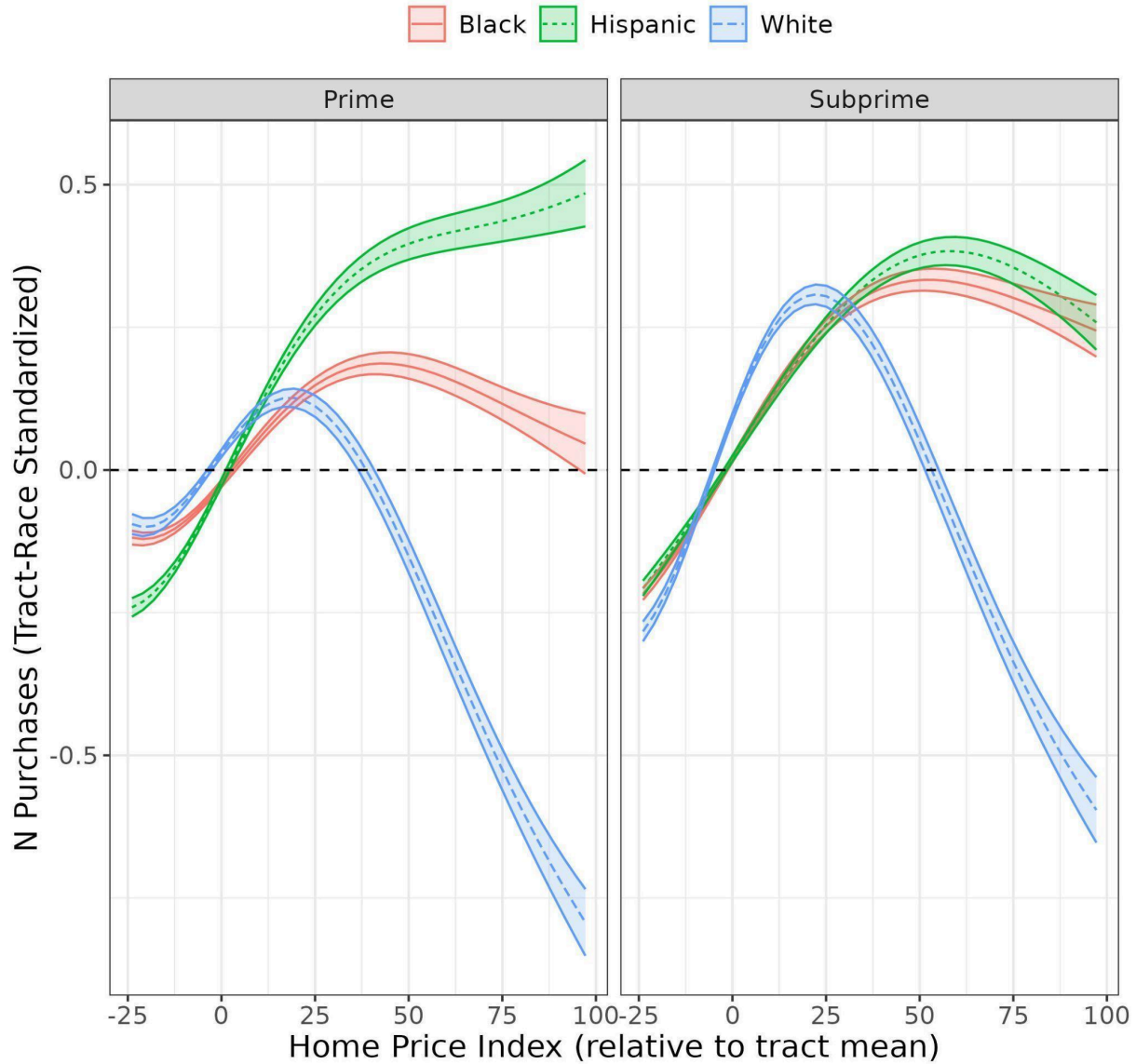
Figure 4: Ethno-Racial Differences in Timing of Mortgage-Funded Home Purchases Across the 2000s U.S. Housing Price Boom



Data Sources: HMDA and U.S. Census

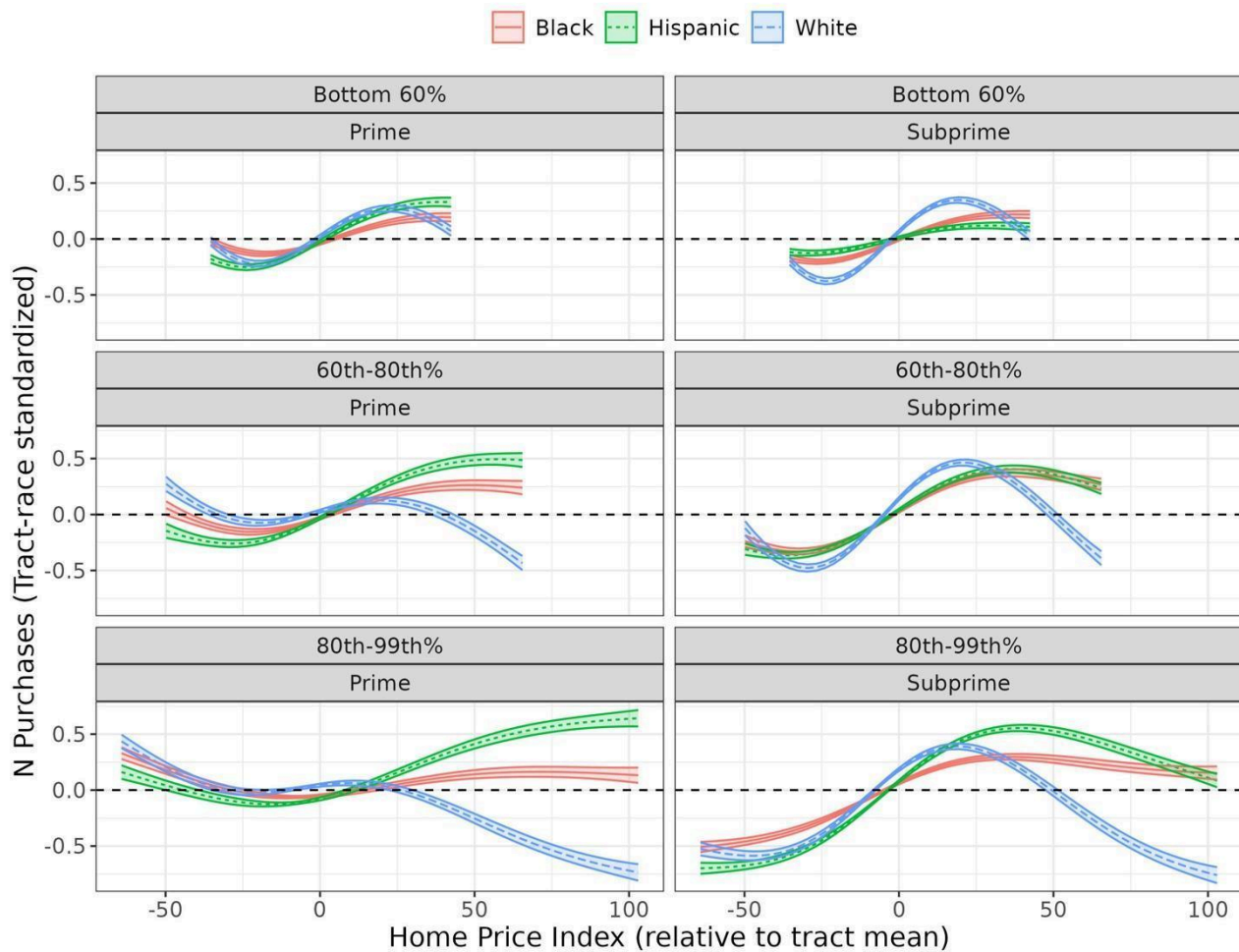
Note: The middle panel of Figure 4 shows each group's share of all first-lien, owner-occupant, home-purchase mortgages originated in a year (HMDA), divided by that group's share of all household heads (U.S. Census). We use Census counts of households rather than total adult population as the denominator in order to account for potential differences in household size. The plotted series include purchases made with both high-cost (i.e. subprime) and conventional loans.

Figure 5: Model-adjusted Relationship Between Neighborhood HPI Appreciation and Race-Specific House Purchase Rates



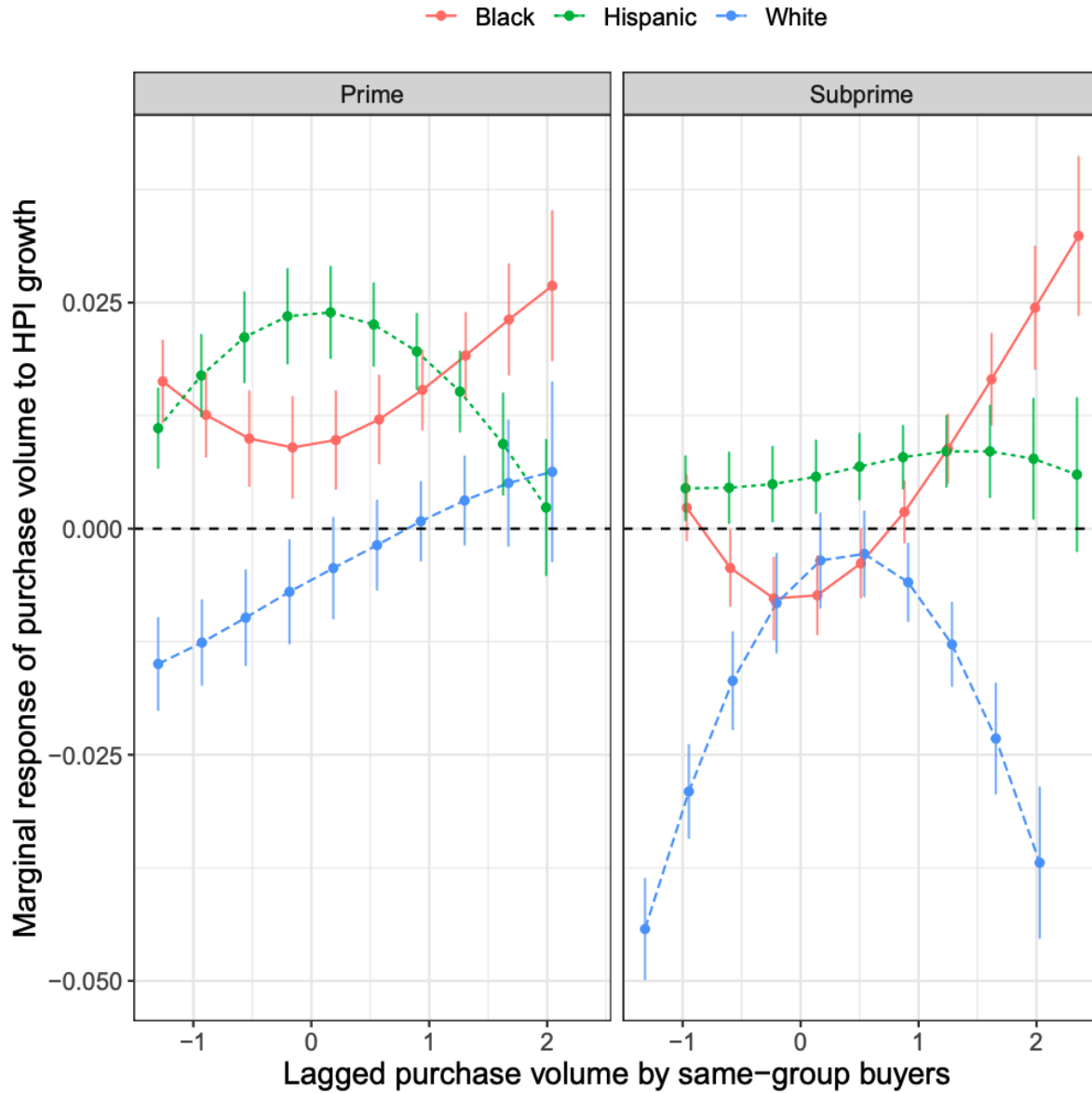
Note: Figure 5 shows GAM model estimates of the number of race-specific purchases in a tract-year as a non-parametric function of the HPI, indexed to 2000. The model controls for metropolitan racial population shares, lagged subprime rate (as a proxy for credit availability), racial homeownership rates (a proxy for buyer pool), and tract fixed effects. We fit separate models for each racial group and mortgage type (prime/subprime). The figure plots predictions across the distribution of tract-level HPI values, with covariates held at observed values. HPI values are plotted relative to their tract-specific means. In both the prime and subprime cases, Black and Hispanic purchase volumes increase with bubble growth, whereas white purchase volumes peak early in the bubble and decline as the bubble inflates.

Figure 6: Varying Association Between Neighborhood HPI Appreciation and Race-Specific House Purchase Rate, by Strata of Low-/Medium-/High-Bubble Intensity Locales



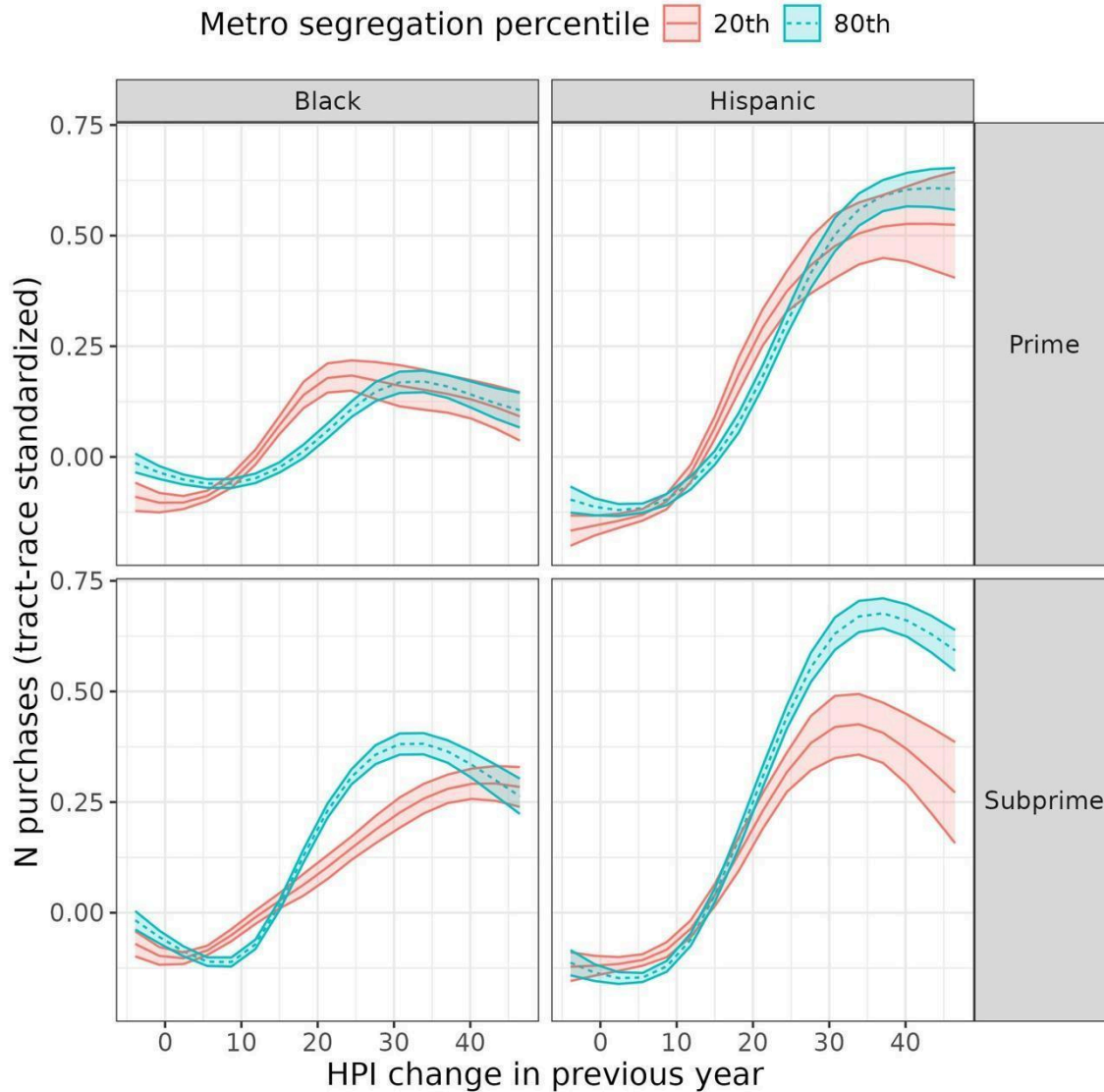
Note: Figure 6 shows estimates of race-specific purchase rates over the HPI distribution, by strata of local bubble intensity. We measure strata of local bubble intensity as the difference between the local peak and trough of home prices (i.e. how hard the local housing market crashed). We then group tracts by percentiles of bubble intensity: bottom 60 percentiles, 60th-80th percentiles, and top 20 percentiles. HPI values are indexed to 2000 and plotted on the x-axis relative to their tract-specific means. The estimates are drawn from GAM models, in which the total number of mortgages originated to each racial group in a tract-year is estimated as a non-parametric function of a bubble intensity stratum-home price interaction, with controls for metropolitan racial population shares, lagged subprime rate (as a proxy for credit availability), racial homeownership rates (a proxy for buyer pool), and tract fixed effects.

Figure 7: Marginal Effect Plot of HPI Growth on Racial Group Market Share, by Growth in Same-Group Purchase Share in Prior Year



Note: Figure 7 plots the estimated average marginal effect of price increases on home purchases over the distribution of lagged same-group entry rates. The estimates are drawn from GAM models, in which the total number of mortgage originations originated to each racial group in a neighborhood is estimated as a non-parametric function of the interaction between home prices and the lagged number of group-specific purchases within the tract, with controls for metropolitan racial population shares, lagged subprime rate (as a proxy for credit availability), racial homeownership rates (a proxy for buyer pool), and tract fixed effects.

Figure 8: Marginal Effect of HPI Growth on Race-Specific Purchase Share by Racial Segregation



Note: Figure 8 shows estimates of race-specific purchase rates over the HPI distribution, by varying levels of metropolitan racial segregation. HPI is indexed to 2000. The estimates are drawn from GAM models, in which the total number of mortgages originated to each racial group in a tract-year is estimated as a non-parametric function of the interaction between home prices and the (CBSA) dissimilarity index between whites and the respective minority group, with controls for metropolitan racial population shares, lagged subprime rate (as a proxy for credit availability), racial homeownership rates (a proxy for buyer pool), and CBSA fixed effects.